A Framework for Cryptographic Verifiability of End-to-End AI Pipelines

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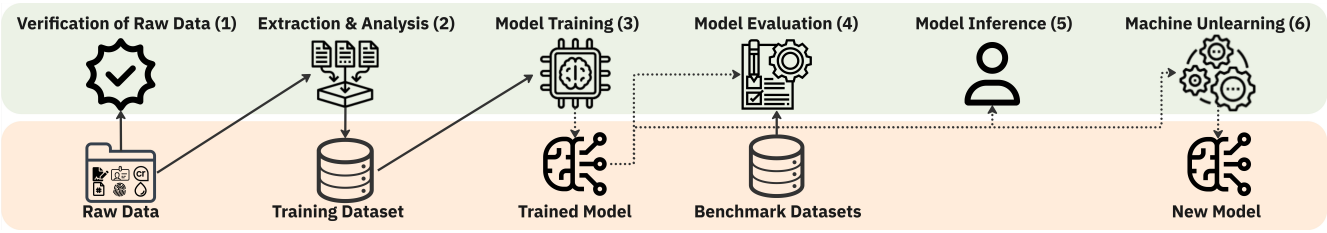


Figure 1: Framework for end-to-end verifiable AI pipelines: top-row icons represent Zero-Knowledge Proofs for process integrity; bottom-row icons represent Cryptographic Commitments to data objects. Solid/dashed arrows indicate data/model inputs & outputs of ZKPs, respectively.

Abstract

The increasing integration of Artificial Intelligence across sectors necessitates robust mechanisms for ensuring transparency, trust, and auditability of its development and deployment. This is particularly important in light of recent calls in various jurisdictions to introduce regulation on AI safety. We propose a framework for complete verifiable AI pipelines, identifying key components and analysing existing cryptographic approaches that contribute to verifiability across different stages of the AI lifecycle, from data sourcing to training, inference, and unlearning. This framework could be used to combat misinformation by providing cryptographic proofs alongside AI-generated assets to allow downstream verification of their provenance and correctness. Our findings underscore the importance of ongoing research to develop cryptographic tools that are not only efficient for isolated AI processes, but that are efficiently 'linkable' across different processes within the AI pipeline, to support the development of end-to-end verifiable AI technologies.

CCS Concepts

• **Social and professional topics** → *Technology audits*; • **Computing methodologies** → *Multi-agent systems*; **Artificial intelligence**; • **Theory of computation** → *Data provenance*; • **Security and privacy** → *Social aspects of security and privacy*; *Cryptography*.

*This work was carried out while Kar Balan was an intern at Digital Catapult and was partly supported by UKRI Grant EP/T022485/1.

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IWSPA '25, June 6, 2025, Pittsburgh, PA, USA
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ACM ISBN 979-8-4007-1501-3/2025/06
https://doi.org/10.1145/3716815.3729011

Keywords

Zero-Knowledge Proofs, Verifiability, Artificial Intelligence, Machine Unlearning, Data Provenance, AI Compliance

ACM Reference Format:
Kar Balan, Robert Learney, and Tim Wood. 2025. A Framework for Cryptographic Verifiability of End-to-End AI Pipelines. In *Proceedings of the 2025 ACM International Workshop on Security and Privacy Analytics (IWSPA '25), June 6, 2025, Pittsburgh, PA, USA.* ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3716815.3729011

1 Introduction

The rapid adoption of Artificial Intelligence (AI) across diverse sectors has raised significant concerns around the transparency, trust, and auditability of such systems. As AI models become increasingly embedded in critical decision-making, the inability to verify the authenticity, provenance, and operational behaviour of these models poses critical challenges. The increasing use of AI value chains—where data, models, and results are exchanged across various stakeholders [35]—creates a need for robust mechanisms and pipelines that ensure trust through verifiability.

A key scenario illustrating these challenges is that of outsourced model training, where it is important to verify that models are trained using the specified parameters, processes, and datasets [17]. Another challenge arises when different versions of a model are served via opaque APIs: users have no means to verify which model is being served, and may be using one that is cheaper for the server to run rather than the one they paid for [58]. It may not be immediately easy to distinguish between model versions: model quality (e.g. accuracy) may be hard to detect manually.

These challenges are becoming increasingly relevant in the present context of evolving regulatory initiatives, most notably the European Union's AI Act [44], which demands increased transparency and auditability of AI systems, as well as guarantees related

to model performance [51]. These regulations often mandate third-party audits and the comprehensive review of technical documentation. At the same time, the AI industry is focused on safeguarding the privacy of their models and sensitive data. This trade-off between regulatory demands for openness and the industry's desire for privacy motivates developing technical solutions to enable trustless audits and verifiability of AI systems without compromising the privacy of underlying data, processes and models [54]. Cryptographic verifiability offers a high level of automatable assurance and provides a way to achieve strong guarantees of the correctness and integrity of various aspects of AI verifiability:

- Digital signatures and cryptographic commitments are used in the C2PA framework (discussed in §2.2.2) to provide assurance of media provenance. Cryptographic key material can be held in tamper-proof hardware (e.g. chips within digital cameras) to increase the confidence that a signed image was produced by a particular hardware device.
- Zero-knowledge proofs (ZKPs) in various forms allow proving various model properties, including correctness of model training, data provenance and integrity, or execution of model inference, to third parties without revealing inputs they may wish to keep private (such as model weights).
- Federated Learning (FL) allows models to be trained without individual entities revealing their data to a third party. This was notably used by Google to improve autocorrect on Android keyboards while maintaining privacy of individual user inputs [37].

The potential for an end-to-end verifiable AI pipeline is only possible through recent advancements in cryptography and verifiable computation due to the high computational complexity of AI training and inference. However, as yet, there are no implementations of this fully verifiable pipeline, although work towards this goal has begun, e.g. [54]. In this paper, we propose a framework for achieving end-to-end verifiability in AI pipelines. Specifically, we provide the following contributions:

(1) **A framework for end-to-end verifiable AI pipelines** (§4): We introduce and outline a framework for verifiability across the AI pipeline, defining the key components required. We review existing solutions and techniques for verifiabality in different parts the AI pipeline—from data provenance and model training, to model evaluation and machine unlearning, and map these approaches onto our framework.
(2) **Gap analysis** (§5): We conduct an analysis to identify the gaps and challenges in achieving full end-to-end verifiability within the AI pipeline. Building on the mapping of existing techniques to our framework, we also propose (at a high level) a generic method for connecting parts of the pipeline together and highlight critical areas where current techniques fall short of achieving full end-to-end verifiability.

We hope that our framework, review of current solutions, and gap analysis motivate further research to develop the necessary tools and technologies to secure the entire AI pipeline.

2 Preliminaries

In this section we give an overview of the cryptographic tools used for AI pipeline verification, which form the technical basis for many of the components in our proposed framework. We also discuss existing methodologies for using these tools in a standardised way.

2.1 Cryptographic Tools

Most cryptographic protocols discussed in this paper involve the use of many cryptographic primitives. In this subsection, we describe three fundamental primitives: digital signatures, cryptographic commitments, and Zero-Knowledge Proofs (ZKPs). We refer the reader to the literature for technical definitions.

2.1.1 Digital signatures. Digital signatures are ubiquitous in digital infrastructure, notably in X.509 certificates. They are used to ensure data is from an authentic source and has not been tampered with in transit [45]. There are many security definitions for digital signature schemes. A common security goal is Existential Unforgeability under Chosen Message Attack (EUF-CMA): given a number of message/signature pairs for a fixed public key, it should be computationally infeasible for an adversary to construct a signature for a new message that verifies correctly.

2.1.2 Cryptographic Commitments. A cryptographic commitment scheme is a mechanism for 'committing' to a message (e.g. a statement or dataset) without revealing it. A party will create a cryptographic commitment to send to another party, or to post publicly, so that they can 'open' it later and reveal the message. This allows them to declare they know certain information before some event occurs, or to prove their message is not dependent on some external information that has not yet been revealed, or simply to make a declaration that cannot be changed once made public. Security in this context means (informally) that the commitment reveals nothing about the information committed to ('*hiding*'), and that the commitment cannot be opened to a different value ('*binding*').

Cryptographic commitments are useful in the context of AI pipelines because they can be combined with ZKPs (discussed later) to prove statements about a model, e.g. that a particular inference output was computed using a specific model without revealing model weights.

2.1.3 Zero-Knowledge Proofs (ZKPs). We give a brief, high-level summary of ZKPs in this section. The reader is referred to [10, 19] for a formal treatment.

A ZKP is a protocol between a prover and a verifier in which the prover tries to convince the verifier that some statement is true while revealing no information other than this fact. They can be 'non-interactive', where the prover can publish a proof for anyone to verify offline. There are many variants, aiming to improve different aspects of the protocol (e.g. the amount of prover or verifier computation required, or the size of the proofs, or assumptions on the adversary's capabilities).

Proofs that do not offer the strong privacy guarantee that it is only possible to learn whether not the statement is true (and not some extra information about the prover's inputs) can be useful for demonstrating that a public computation was performed honestly without having to re-compute it, which may be costly [24].

Well-known proof systems include Succinct Non-interactive Arguments (SNArgs) [38], Succinct Non-interactive Arguments of Knowledge (SNARKs) [24] and Succinct, Transparent Arguments

of Knowledge (STARKs) [9]. They are prefixed with *zk-* when they also satisfy the zero-knowledge property (e.g. zk-STARK).

2.2 Authentication Standards

2.2.1 Cryptographic Infrastructure. Cryptographic standards are important for ensuring that different low-level implementations are interoperable. Digital signature schemes have been standardised in many places, e.g. by NIST and ISO. By contrast, ZKPs have only recently seen adoption in industry, so there is currently no widely adopted standard, although IETF drafts have been written to attempt to standardise approaches [25]. ZKProof (https://zkproof.org/) exists to support standardisation of ZKPs.

Standards for cryptographic 'infrastructure' go beyond the low-level primitives to provide a common way of communicating cryptographic artefacts regardless of the algorithm specifics. For example, ECDSA and RSA digital signatures can be used interchangeably in X.509 Public Key Infrastructure (PKI). To the authors' knowledge, there are currently no standards for infrastructure supporting the distribution of ZKPs except those confined within specific ecosystems (such as in certain blockchains).

2.2.2 Content Credentials Metadata. The Coalition for Content Provenance and Authenticity is the cross-industry standards group that developed C2PA - an open specification for encoding provenance metadata within media and other binary assets [4]. C2PA encodes data provenance through a data structure called a 'manifest' which contains information, referred to as 'assertions', detailing the asset's origin, including who created it, how, where, when, as well as the source assets ('ingredients') used. Should the ingredient assets also bear C2PA manifests, a graph is created, containing the provenance information of all ingredients. Multiple assertions are grouped within a claim and cryptographically signed with keys authenticated using standard PKI. C2PA securely creates bindings between the manifest and asset. Hard bindings rely on cryptographic hashes to ensure that the manifest belongs to a specific asset and that the asset has not been altered. For images, this involves hashing pixel data or specific file structures, while for non-media assets, it involves hashing raw data or defined byte ranges. Soft bindings use techniques such as content fingerprints or invisible watermarks to identify assets even after transformations or format changes which invalidate hard bindings. The ontology of assertions is flexible and can be extended by the community, containing flags such as `data_mining`, `ai_inference`, and `ai_training`, which can empower creatives to specify training consent for their digital assets.

Content credentials are increasingly being adopted across various industries. C2PA manifests are now automatically generated and embedded by several camera manufacturers [50] and creative tools [11]; generative models, such as Adobe Firefly and OpenAI DALL-E, are automatically creating C2PA manifests for their generated assets, marking them as synthetically generated [43].

3 Framework for End-to-End Verifiable AI Pipelines

In this work, we propose our *framework for end-to-end verifiable AI pipelines*. The idea is to verify (zero-knowledge) proofs that each process in the pipeline has been executed honestly, and that inputs to each process match public cryptographic commitments to the outputs of the previous process. Linking ZKPs together using commitments is already being done for some parts of the pipeline, as discussed in section 4. The breakdown of the pipeline is shown in fig. 1 and summarised below:

1. Verification of raw dataset: Verifying that a corpus of raw inputs, such as images and text, that has been compiled as the initial dataset, contains authentic assets (e.g. by checking digital signatures). Additional properties, such as dataset bias or representativeness, may also be assessed and verified during this phase through Zero-Knowledge Proofs of Fairness (ZKPoFs).

2. Extraction & Analysis: Collating, processing, and analysing data from the raw dataset to compile the final training dataset. Transformations such as normalisation or conversion to tensor format are applied at this stage to prepare for training. Current pipelines fail to provide full verifiability during this phase.

3. Model Training: Training the model using the training dataset from above as input. Recent advances have made it feasible to apply cryptographic proofs to training, particularly addressing privacy concerns during audits. Zero-Knowledge Proofs of Training (ZKPoTs) enable stakeholders to verify that training was conducted correctly using a specific dataset and algorithm, providing confidence in the model's development process.

4. Model Evaluation: Assessing whether a model satisfies its design requirements. Proofs can be generated to confirm model performance on benchmark datasets, relevant for regulatory contexts like the AI Act. These proofs allow for the private aggregation of model performance metrics, ensuring verifiable evaluations without revealing sensitive data. ZKPoFs may be used here.

5. Model Inference: Deploying and serving the model to users either directly (e.g. publishing model weights) or via API access to the model running on a server. Zero-Knowledge Proofs of Inference (ZKPoIs) enable a party to prove that an AI model has made a correct inference using a specific, previously committed to model, without disclosing the model weights.

6. Machine Unlearning: Verifying removal of specific training data points from AI models without retraining from scratch or disclosing sensitive information. Zero-Knowledge Proofs of Unlearning (ZKPoUs) enable proving this without revealing what data was removed (although it is often required that this be known, so the zero-knowledge property may not be needed). This is crucial for compliance with data privacy laws like the GDPR.

Recent focus on AI verifiability has been on providing assurance over isolated parts of this pipeline: e.g., ZKPoTs, discussed in §4.2, focus solely on proving that the training process was executed honestly given a set of inputs. However, to the authors' knowledge no work has yet achieved a full end-to-end verifiable pipeline. Such a pipeline would enable an end user who obtains some model inference to verify cryptographically that the output they received was honestly computed using a specific model, and that this model was trained according to a specific algorithm on a specific dataset.

Not all use cases demand a fully verifiable AI pipeline, such as in informal contexts. In some use cases, it may be important for users to be able to verify one or two steps in the pipeline. In security-critical or commercial settings, users may need assurance both that the model was trained honestly from a set of (possibly public) inputs, and that inference was honestly evaluated by the model on

their input. A high level of assurance can be achieved by 'connecting' proofs for different parts of the AI pipeline cryptographically. This challenge motivates the study and development of a comprehensive framework for an end-to-end cryptographically-verifiable AI pipeline, which is the central focus of this paper.

We see significant potential for regulatory oversight and compliance frameworks to leverage verifiable AI pipelines. Regulatory authorities, such as those under the AI Act, could play a key role in verifying cryptographic proofs across the pipeline to ensure compliance with legal and ethical standards. Integrating our verifiability framework into regulatory standards could help align AI innovation with public interest and legal safeguards.

In §4, we map existing techniques for verifying components the AI pipeline onto our proposed framework. In §5, we discuss the key gaps and challenges preventing full end-to-end verifiability.

4 Mapping Current Tooling to the Framework

We now explore the key technologies and tools currently available for achieving verifiability across the end-to-end AI pipeline. We review existing solutions and techniques that contribute to verifying various stages, from data provenance and model training to inference and unlearning and map these tools onto the components of the proposed framework, highlighting the current state of the art in AI pipeline verifiability.

4.1 Data and Transformation Verification Mechanisms (Framework Steps 1 & 2)

In the AI pipeline, data provenance refers to the detailed record of the origin of data and transformations made on it before being ingested by training algorithms. It is generally achieved using metadata, fingerprinting, and watermarking, which have been described as the 'three pillars of data provenance' [15].

Verifiable data provenance has become critical in the modern era as digital content manipulation and generation technologies continually improve. It is particularly important in the context of AI training to ensure models are unbiased and robust. Without verifiable provenance, there is a risk of training AI systems on manipulated, biased, erroneous, or copyrighted data without creator consent, which can lead to serious privacy breaches [13], the spread of misinformation, and generation of non-consensual imagery [34].

One key technology aimed at verifying the authenticity of digital images and their provenance is the C2PA standard described in §2.2.2, which allows cameras to digitally sign each photo alongside important metadata such as time and location. However, this approach has some limitations. If an image undergoes any form of editing—such as cropping, resizing, or applying filters—the original camera signature no longer corresponds to the edited image and cannot be used to verify the authenticity of the modified version. C2PA-enabled editing software can re-sign the edited image, but this presumes trust in the editing software. If a malicious actor compromises this software, they could forge signatures on any image, undermining the trust in the system. C2PA supports integration with Self-Sovereign Identity (SSI) and Verifiable Credentials (VCs), which are mechanisms for establishing identity in a decentralised and secure manner. By leveraging decentralised identifiers (DIDs), VCs enable entities involved in creating or modifying digital assets to issue cryptographically verifiable claims about their identity. These claims, attached to the content, provide assurances about the origin and authenticity of the asset.

Balan et al. propose DECORAIT [7], a decentralised tamper-evident registry for storing C2PA manifests using blockchain technology. DECORAIT enables the retrieval of provenance metadata for images even if the original metadata has been stripped or if the images have undergone visual editing or transformations using robust image fingerprinting. The system ensures that provenance information remains accessible despite metadata removal, eliminating the need for trust in a centrally-managed manifest registry.

ZKPs allow images to be verifiably transformed without revealing the original image or requiring trust in editing software. PhotoProof [41] uses zk-SNARKs to prove that an image which has undergone transformations such as cropping, flipping, transpose, rotation and brightness adjustment, corresponds to an original image that is authenticated with a digital signature computed in secure hardware. However, its key and proof generation times (367 and 306 seconds for 128x128 images) make it impractical for real-world deployment. Kang et al. propose ZK-IMG [29], a library that achieves a 100x speed-up over PhotoProof, handling HD image transformations efficiently on commodity hardware. Ko et al. [31] introduce an efficient verifiable UHD image redacting scheme using SNARKS with proving and verification times under one second.

Datta et al. [16] propose VerITAS - a system combining zk-SNARKs with the C2PA framework to verify that permissible image transformations, such as cropping or resizing are correctly applied to C2PA-signed authentic images [6]. These proofs integrate with the C2PA standard, ensuring that the image's provenance metadata remains verifiable even after transformations, thus preserving authenticity without exposing sensitive content.

While research and technological progress have been predominantly concentrated on image transformation verification, similar techniques can be applied to audio data. Kang et al. [28] propose a method using attested microphones and zk-SNARKs to ensure the authenticity of audio recordings and transformations. Although attested microphones, which cryptographically sign audio to verify its source, are not yet commercially available, Kang et al. simulate this functionality using cryptographic keystores ('wallets') on the Ethereum (https://ethereum.org/) blockchain to sign audio recordings. zk-SNARKs are then employed to validate the authenticity of the audio edits while preserving the privacy of the original content.

Towards verifiability of unbiased ('fair') training, OATH [21] provides a framework for allowing users to confirm inference outputs come from models trained fairly with respect to various 'demographic variables' (e.g. sex, race, location). Similarly, Confidential-PROFITT [48] proves that decision trees were trained fairly, starting from commitments to trained models.

4.2 Zero-Knowledge Proof of Training (Framework Step 3)

Zero-Knowledge Proofs of Training (ZKPoTs) enable a model trainer to demonstrate that a ML model was trained according to some public algorithm. Traditionally, ZKPoTs have been considered computationally prohibitive due to the complexity and scale of modern ML models, and until very recently it was deemed impractical to

apply ZKP systems to large-scale neural network training [5]. However, recent advances in ZKP techniques have made ZKPoT feasible for useful, albeit limited, models.

There are various options regarding privacy in ZKPoTs: training data could be kept as a private input with corresponding cryptographic commitments as a public input so that the data itself need not be revealed when verifying the proof. This could improve privacy during audits: model providers could respond to audit requests by privately computing any function of the dataset (or model) and releasing its output alongside another zero-knowledge proof certifying the correct execution of the function [54]. Similarly, the ZKPoT could assert that some public commitment is indeed a cryptographic commitment to a model trained on particular data according to a specific algorithm, without directly revealing the model weights. ZKPoTs are therefore relevant in scenarios where ML models are trained by a party that needs to provide assurances about their training process in regulatory contexts, such as the EU AI Act, to ensure the integrity and correctness of the training process while maintaining data and model privacy. Notably, ZKPoTs do not hide the training process (which is often a trade secret) since it is essentially encoded in the (public) proof and/or verification algorithm.

There are still relatively few publications specifically addressing proofs of training; we review them in this section. We note that the majority of early ZKPoT papers focused on simple models, such as logistic regression and decision trees, whereas real-world applications typically demand more sophisticated models like Convolutional Neural Networks (CNNs) and Deep Neural Networks (DNNs). Our focus is therefore on works that extend ZKP capabilities to more complex architectures. We also review works on ZKPs in fine tuning (§4.2.1) and federated learning (§4.2.2), which are both part of the training process in the AI pipeline.

Xing et al. [61] explore verifiable execution, training, and inference in machine learning, including methods for outsourced machine learning. However, many approaches are not zero-knowledge, as they assume the inputs and model are fully visible to both the prover and verifier, making them unsuitable for end-to-end pipelines where privacy is necessary. Their survey identifies two key challenges in this research area: generalisability and efficiency. Generalisability refers to the fact that, in many ZKP systems, floating-point computations in ML algorithms can only be simulated via arithmetic in large finite fields, which is often either costly or induces precision loss. Efficiency challenges arise from the fact that machine learning tasks are inherently computationally expensive, and existing ZKP systems are typically slow and memory-intensive even for much simpler computations. Indeed, even the most efficient generic ZKP systems are insufficient for proving correctness of circuits as large as the training process of a ML model, without significant optimisation [5]. In order to address these challenges, researchers have explored various optimisation strategies. The survey categorises the approaches into three groups: optimising computation processes to reduce complexity, using polynomial approximations (e.g., for sigmoid and ReLU) to minimise verification costs at the expense of some precision, and directly representing activation functions through more complex operations.

zkMLaaS [26] and VeriML [66] are privacy-preserving frameworks for Machine Learning as a Service (MLaaS) that provide probabilistic rather than full verification of model training through challenge-response protocols and random sampling to generate proofs for specific training epochs and iterations. zkMLaaS maintains model accuracy with only 1%–4% degradation resulting from data quantisation and non-linear function approximation. VeriML ensures correctness, fair payment, and trustworthy resource accounting via blockchain-based smart contracts. VeriML supports six ML algorithms and incorporates optimisations such as the Remez approximation and quadratic ReLU approximations. For the CNN model trained on the MNIST dataset with a batch size of 128, VeriML reports a proof generation time of 16.7 s and a verification time of 1.231 s, compared to a native execution time of 1.523 s.

zkDL [52] is a ZKP system for deep learning with two main contributions: 1) zkReLU—a specialised ZKP for the ReLU activation and backpropagation which captures non-arithmetic relationships by leveraging auxiliary inputs to reconstruct intermediate tensors; 2) FAC4DNN—converts deep learning into an arithmetic circuit, efficiently aggregating proofs across layers and steps. Using GPU parallelisation via CUDA, zkDL significantly improves proof times, generating proofs in under a second per batch update for an 8-layer, 10M-parameter network with batch size 64.

ZKAUDIT [54] is a ZKP-based cryptographic protocol for trustless audits of deep neural networks, preserving the privacy of model weights and training data. The protocol has two phases: ZKAUDIT-T, for generating ZKPoTs, and ZKAUDIT-I, for proving inference results (see §4.3) or other audit functions. The paper makes significant contributions in creating efficient ZKPs for stochastic gradient descent on modern neural networks, implementing optimisations like rounded division in finite fields and high-performance softmax operations. Audit scenarios explored include copyright audits, censorship detection, and counterfactual audits, with costs of running ZKAUDIT ranging from $108 for a copyright audit to $8,456 for a comprehensive counterfactual audit.

Abbaszadeh et al. propose KAIZEN [5], a novel approach to ZKPoTs tailored to deep neural networks. It is designed to handle multiple iterations through a recursive structure, allowing the prover to efficiently validate the execution of gradient descent. By leveraging a recursive sumcheck protocol, KAIZEN amortises the proving costs over multiple training iterations, significantly reducing the overhead typically associated with each iteration.

4.2.1 Fine-Tuning (Framework Step 3). Fine-tuning adapts pre-trained models to specific tasks by training on task-specific datasets [27]. It is central to AI development, particularly as foundational models are increasingly sold or licensed for downstream adaptation. Fine-tuned open-source models often outperform proprietary generalist models in narrow tasks [65], driving widespread fine-tuning across various industries. This raises regulatory implications, as the parties performing substantial fine-tuning may be considered providers, rather than deployers of AI systems under the EU AI Act [44]. Verifiability must therefore extend to fine-tuning to support regulatory compliance throughout the AI lifecycle.

Fine-tuning methods can be broadly categorised into full fine-tuning, which updates all model parameters, and parameter-efficient fine-tuning (PEFT), which updates a small subset of the model's parameters [65]. The technical requirements for ZKPs of fine-tuning largely align with those for pre-training. For full fine-tuning, existing ZKPoT schemes can be adapted with minor adjustments to

accommodate the new dataset and weights initialisation. However, for PEFT, separate circuits must be designed to validate the specific operations performed, such as updating only a subset of weights.

Garg et al. [23] introduce the first work to explicitly address the concept of zero-knowledge proofs of fine-tuning. The paper does not provide specific technical implementation details of the proposed ZKP system for fine-tuning, but does propose methods to ensure the integrity of inputs from foundational models. For instance, the authors suggest using digital signatures to verify the provenance of the committed inputs. Moreover, the system requires the prover to demonstrate not only that the model was trained honestly but also that the input data is well-formed, proving the prover's knowledge of valid signatures for the committed inputs.

vTune [64] offers an alternative approach to fine-tuning verification in which backdoors are embedded into training data to create subtle but detectable signals in model outputs. While their method is efficient compared to producing ZKPs, it does not offer the strong assurance of cryptographic verifiability, which is our goal.

4.2.2 Federated Learning (Framework Steps 1, 2 & 3). Federated Learning (FL) is a type of distributed machine learning that enables multiple participants to collaboratively train a model without sharing their raw training data with a central server. Typically, clients train the model locally on their devices and only share model updates (e.g., gradients or weights) with a central aggregator; however, aggregator-free systems have also been proposed [8]. FL enhances training data privacy, however, this introduces additional challenges in verifying the authenticity of client-provided model updates and ensuring the security of the training and aggregation processes.

Regulatory frameworks, such as the EU AI Act, apply to FL participants, who may be considered 'providers' under certain circumstances, subjecting them to additional legal responsibilities, including ensuring compliance with data protection regulations and maintaining transparency in their processes. Thus, the ability to provide proof and be subject to auditing is crucial for compliance and accountability. Incorporating ZKPs naturally complements this, enabling proof of correctness in a privacy-preserving manner.

FL participants could be required to commit to their data publicly to enable the auditing of model inputs, disincentivising dataset manipulation to poison models [36]. In addition, ZKPs could be used to prove that a client has followed the expected learning algorithm, or that their training data satisfies certain predicates (e.g. numerical values lie within certain bounds), ensuring consistency and fairness across participants. In systems with a central aggregator, ZKPs can be employed to guarantee that the aggregation of model updates was performed correctly and without tampering [57]. Another relevant use case for ZKPs is proving ownership in FL scenarios where multiple stakeholders contribute to the development of a model. This could play a vital role in managing rights, rewards and responsibilities within collaborative AI ecosystems, helping to maintain trust among participants.

Ruckel et al. [46] provide a proof of concept FL implementation involving multiple linear regression models, which allows clients to prove that they have truthfully trained their model updates based on commitments to private data. The system can incorporate proofs of provenance, ensuring that data originated from a certified source, such as an attested sensor as discussed in §4.1.

Wang et al. introduce PTDFL [56], a decentralised, aggregator-free FL framework. Each node conducts local training on its dataset and generates a ZKPoT to validate the correctness of the computed gradient. This proof, along with the encrypted gradient ciphertext, is broadcast to the other participating nodes. Each node verifies these proofs before aggregating the gradient ciphertexts, ensuring that only trustworthy model updates contribute to the global model.

martFL [32] is an FL framework that facilitates a secure, utility-driven data marketplace for AI models. It enables the 'data acquirer' (model buyer) to prove in zero-knowledge that it has accurately aggregated local models according to pre-committed weights and ensures that 'data providers' (model trainers) can claim rewards that are proportionate to their contributions.

Flais et al. [60] discuss the implications of the EU AI Act for FL. Although all parties bear some degree of liability under the AI Act, the authors propose a strategic shift to designate the server operator as the primary service provider. One significant challenge arises from the stipulation that any data involved in training processes must be evaluated for potential bias or adversarial information, which poses difficulties for the server, as the training data resides within local nodes, limiting its ability to conduct thorough evaluations. Here, ZKPs can support compliance by providing cryptographic assurances that model updates are derived from previously committed training data, enabling clients to demonstrate the integrity of their contributions without necessarily exposing sensitive information.

In summary, the integration of ZKPoTs has the potential to enhance privacy, auditability, and compliance in both centralised and distributed AI. By leveraging ZKPs, systems can mitigate risks related to data manipulation and model poisoning, thereby enhancing the security and accountability of collaborative AI ecosystems. However, significant limitations exist—notably the inability to keep the model architecture hidden from the verifier when checking a ZKP. This presents challenges in scenarios where model architecture confidentiality is essential. One possibility for addressing this concern is for regulators to verify proofs privately and issue certificates to service providers instead of proofs being released publicly. Despite this, ZKPs still allow for the privacy of the resulting weights, as well as input data. Additionally, while ZKPoTs ensure honest training by preventing data duplication and manipulation, they fall short of providing comprehensive guarantees of model correctness—there remain opportunities for biases. Research focusing on how ZKPs might be combined with alternative model fairness checking methods to enhance guarantees of correctness and mitigate potential biases would be beneficial.

4.3 Zero-Knowledge Proof of Inference (Framework Steps 4 and 5)

Zero-Knowledge Proofs of Inference (ZKPoIs) allow one party (the prover) to demonstrate to another party (the verifier) that a specific machine learning model has made a correct inference on a given input, while keeping the input or the model (or both) private. This ensures the privacy of both the data and the model, which can be particularly important in sensitive applications such as healthcare and security. The ability of ZKPoIs to prove the accuracy of model predictions without exposing the underlying model weights is especially relevant in regulatory contexts like the AI Act, which

mandates ensuring model accuracy and robustness. ZKPs can be generated during model inference to produce proofs of the model's outputs on benchmark datasets, allowing for verifiable attestations that confirm the model's accuracy and performance [51]. Another important use case involves verifying that the correct version of a model is being served, particularly in cases where users pay for API access without visibility of the underlying model. ZKPs further enable trustless verification in scenarios like legal subpoenas, allowing a responder to use an ML model to privately retrieve and verify documents that match a request, without exposing the underlying data or tampering with the results [14]. Additionally, as AI models are increasingly involved in decision-making processes, proofs of inference enable establishing a verifiable chain of provenance for decisions made using AI models. Two surveys [39, 61] provide analyses of various ZKP systems, offering insights into the trade-offs between performance, scalability, and privacy.

The Modulus Labs whitepaper [39] benchmarks multi-layer perceptron (MLP) architectures with increasing parameter counts (up to 18M parameters) and network depth (up to 500 layers) against a set of ZKP systems for inference, including Groth16 (https://github.com/arkworks-rs/groth16), Gemini [12], halo2 (https://zcash.github.io/halo2), Plonky2 (https://github.com/0xPolygonZero/plonky2), Winterfell (https://github.com/facebook/winterfell) and zkCNN [33]. The analysis highlights Plonky2 as the best performing in terms of proof generation time. For 15M parameters, proof generation times vary significantly, with halo2 taking around 275 seconds compared to Plonky2's 50 seconds. However, Plonky2's memory consumption is notably higher, with a network of 11M parameters requiring 42GB of memory compared to halo2's 19GB. As the model size increases to 18M parameters, the memory usage gap narrows, with halo2 consuming 48GB and Plonky2 56GB. zkCNN performs best in handling larger models, achieving top two results across experiments for both proof generation time and memory efficiency.

Xing et al. [61] review developments in verifiable inference and execution up to June 2023, emphasising the key challenges of generalisability and efficiency, discussed previously in the context of ZKPoTs (§4.2), which apply equally to verifiable inference.

Kang et al. [30] presented the first ZKP scheme for ImageNet-scale MobileNet v2 using the halo2 framework to create an efficient zk-SNARK circuit, unlike prior work which focused on significantly smaller models. They optimised division verification with custom gates for linear layers and a lookup argument for non-linear layers, while ensuring privacy through hash commitments. Building on previous work, Kang et al. [14] introduce ZKML, a framework designed to generate zk-SNARKs for a range of modern ML models, including large-scale models like GPT-2 and Stable Diffusion, unlike prior schemes, which were limited to CNNs. The accuracy of models such as VGG16 and ResNet-18 differed by only 0.01% from their floating-point counterparts, demonstrating minimal performance degradation despite the quantisation and arithmetisation changes required for zk-SNARKs.

ezkl (https://ezkl.xyz/), built on halo2, is currently the most mature and comprehensive open-source library for integrating zero-knowledge proofs with ML models, however it is in its early beta stage and is not suitable for use in production environments. It enables verifiable and privacy-preserving inference by automatically converting AI models from ONNX format into zk-SNARK circuits.

South et al. [51] use zk-SNARKs as proofs of model performance—proving that a model achieves specific performance or fairness metrics on public benchmarks. Similarly to ZKML, the model weights are kept private but the model architecture is public. South et al. propose a 'predict, then prove' strategy which allows model inference results to be delivered immediately, while the proof can be generated later. This is useful in contexts where auditability is important, but the proof generation time is less critical. Individual inference proofs can be naïvely bundled into a single package, which is computationally efficient but exposes inference outputs, presenting privacy risks. Alternatively, an aggregation circuit can be employed to validate each proof and consolidate results, adding a layer of security but increasing computational complexity. Another approach involves constructing a custom halo2 circuit that aggregates only essential metrics, such as accuracy scores or confusion matrices, allowing for private data outputs while sharing verifiable performance metrics. Notably, the nanoGPT model exhibits the highest resource consumption, with a proving time of 46 s, verification time of 2.69 s, and a proof size of 0.7 MB.

FairProof [63] uses ZKPs to verify the fairness of ML models. It consists of two main components: a fairness certification algorithm for fully-connected neural networks, which generates a personalised 'fairness certificate' for each data point, and a ZKP-based cryptographic protocol to verify both the proof of inference and the fairness certificate without revealing the model weights, although the model architecture is public. FairProof generates a verifiable fairness certificate in an average of 1.17 minutes per query without parallelisation on an Intel-i9 CPU, with certificate sizes averaging 43.5 KB, and an average verification time of less than 2 seconds.

ZKPoIs provide a clear pathway to enhancing trust in ML systems by simultaneously providing inference verifiability and model privacy, opening the door to broader adoption of ML services where transparency and security are key. However, as previously discussed in §4.2, a fundamental limitation remains: while ZKPoI ensures that sensitive information like model weights and input data remain private, the model architecture itself is revealed when verifying the ZKP. This limitation poses challenges in scenarios where architectural confidentiality is essential. In such cases, a regulator could verify the ZKPs and issue a compliance certificate to service providers. Enabling model architecture privacy may encourage broader industry adoption of cryptographic verifiability of the AI pipeline, making this an important avenue for future research, extending beyond cryptographic primitives to the wider security architecture.

4.4 Zero-Knowledge Proof of Unlearning (Framework Step 6)

Machine unlearning is the process of removing a specific training data sample and its influence from an already trained ML model. A Zero-Knowledge Proof of Unlearning (ZKPoU) addresses the need to verifiably remove data from ML models without revealing, for example, current or revised model weights. This concept is particularly important in the context of regulations like the General Data Protection Regulation (GDPR)[2], California Consumer Privacy Act (CCPA)[3], and Canada's Personal Information Protection and Electronic Documents Act (PIPEDA)[1], all of which enforce the 'right to be forgotten', which endows individuals with the right

to request the removal of their data, which extends to AI models trained using that data. In a recent opinion [20], the European Data Protection Board emphasised that supervisory authorities may impose corrective measures, such as model retraining, in cases where unlawful data processing occurs during AI model development. ZKPoUs enable verifiable removal of unlawfully included training data, ensuring compliance without retraining.

Unlearning methods are broadly categorised into two types: exact and approximate. Exact unlearning aims to fully remove the influence of specific data points from the model by retraining it from scratch [62]. This provides strong guarantees that the data has been removed, but it comes at a significant computational cost. Approximate unlearning focuses on efficiently reducing the influence of certain data points by updating the model's parameters. Although this method does not completely eliminate the influence of the targeted data, it drastically reduces computational and storage requirements, making it viable for real-world applications [62]. Several surveys have explored the topic of machine unlearning, each classifying techniques into distinct taxonomies, e.g. [42].

Eisenhofer et al. [18] present the first ZKP system for verifiable ML and unlearning processes. The system supports exact unlearning, as well as two types of approximate unlearning: amnesiac unlearning, which removes the updates to the model's parameters that were directly computed on the specific data point, and optimisation-based unlearning, in which the loss corresponding to a specific data point is increased and the resulting model parameters are subtracted from the original model. The system uses Spartan [47], a ZKP system which does not require a trusted setup (see §2.1.3. The system generates three core types of proofs: proof of training, which establishes the integrity of the model prior to any unlearning requests; proofs of unlearning, which ensure compliance with unlearning requests and allow the user to verify that their data is correctly removed from the model; and proofs of non-membership, proving that the specific data point is no longer part of the dataset. One limitation of the system is that users must verify every update to the model to ensure that their data has not been re-added, which is often impractical in real-world systems. To mitigate this, regulators or trusted third parties could step in to verify unlearning requests on behalf of users and ensure compliance across the system. There is also no mechanism to verify that the updated model has been redeployed and is the one served to users, leaving a gap in end-to-end pipeline verifiability.

Weng et al. present a framework for proof of unlearning in Machine Learning as a Service (MLaaS) settings [59] using Trusted Execution Environments (TEEs) to maintain the integrity of unlearning processes and data. The system facilitates efficient retraining by updating only the submodels affected by the unlearning request. While TEEs are typically much more efficient than using ZKPs, they are not immune to security vulnerabilities and can be compromised [40], unlike ZKPs whose security is guaranteed cryptographically.

Gao et al. highlight critical challenges in achieving verifiable machine unlearning, demonstrating that even with sophisticated unlearning mechanisms, privacy attacks can exploit both the original and updated models to infer or reconstruct deleted data [22]. This underscores the importance of developing robust unlearning frameworks that not only efficiently manage data removal but also provide strong privacy guarantees.

ZKPoUs could play an important role in ensuring that ML models comply with data privacy regulations and the right to be forgotten by providing individuals with the assurance that their data can and has been removed from AI systems. Despite progress in unlearning methods, significant challenges remain, particularly concerning computational efficiency and the need for robust verification mechanisms. The role of regulators or trusted third parties becomes vital to ensure compliance and facilitate the verification of unlearning requests. By addressing these challenges, ZKPoUs could contribute to regulatory compliance and ethical AI practices.

5 Gaps in verifiability of the AI pipeline

While solutions from §4 typically target verifiability of individual processes in the AI pipeline, few consider cryptographically connecting the outputs of one process to the inputs of the next. As a result, the pipeline—from raw data through to generated assets or model inference results—remains only partially verifiable. As previously discussed, a full end-to-end verifiable AI pipeline would allow an end user who obtains some model inference to verify cryptographically that the output they received was honestly computed using a specific model, and that this model was trained according to a specific algorithm on a specific dataset. Reducing reliance on human-driven processes and instead using cryptography to verify correctness is closer to a zero-trust architecture, which has become a focus of secure system design in recent years.

Proving the correctness of the entire pipeline 'at once' (i.e. in one monolithic proof) is not a good approach for several reasons. Firstly, designing such a system is a significant task and possibly requires rework for each model architecture, training data format, etc.. Secondly, re-proving correctness of training for every inference is an unnecessary overhead. Thirdly, ZKP systems optimised for training will not necessarily be efficient for inference, and vice versa (e.g. where different circuit arithmetisation techniques are used). Instead, the most natural solution would appear to be for ZKPs at each step in the process to include in their scope the proof of correctness commitments to aspects of the model being trained or evaluated. This solution could be seen as viewing the AI pipeline as an incrementally-verifiable computation (IVC) [53], which is concretely realised in works by Abbaszadeh et al. [5] and Bowe et al. in halo2, discussed in the previous section. In practice, this looks like the following: for each process in the pipeline, take all outputs and commitments from the previous process as input, verify the commitments, execute the current process and compute commitments to the new outputs, and generate a proof that the current process and commitment generation were performed honestly.

The proposed framework for an end-to-end verifiable AI pipeline would further enable comprehensive provenance tracing for both models and their generated outputs. This would help combat misinformation by providing verifiable information regarding training data and attesting to the correctness of the training and inference processes. Ultimately, it would enable users to trace the origins of generated assets and the integrity of the processes that produced them, fostering transparency and trust in AI systems.

The lack of end-to-end AI pipeline verification in the literature appears mainly to be due to the nascency of proving AI processes in zero knowledge and the effort involved in designing ZKPs for

the whole pipeline. Indeed, research papers sometimes provide sketches of how a more complete pipeline could be developed from the solution presented (e.g. descriptions of image input commitments in papers on ZKPoT [5]). In the current work, we argue that developments in ZKPs for AI have reached a level of maturity that now warrants investigating the verifiability of the full pipeline, especially in light of emerging regulatory demands. In this section, we outline the main gaps we have identified and discuss how they might be addressed.

5.1 Gap between raw data and training input data

The first gap identified in the verifiable AI pipeline sits between data in its raw form and data in its processed form, as it enters the training step. While verifiable data provenance (§4.1) is a significant first step towards partially closing this gap and securing the entire AI pipeline, alone it is insufficient. There are two parts to this gap: down-selecting a set of training data from a larger set of raw data through refinement and analysis, and transforming the training data for ingestion by training processes.

The data refinement gap is substantial, encompassing many key data processing stages. Current and upcoming regulatory requirements [44] emphasise the need for robust data governance and data management practices, underlining the importance of documenting data collection processes, the provenance of the data, and the intended purposes for which the data was collected. In many cases, dataset refinement—such as removing NSFW or copyrighted images via automated tools like AI—may be required. Proving the correctness of such processes could itself involve generating and aggregating ZKPoIs of AI-based filtering. This refinement step could ensure compliance with regulatory requirements for dataset preparation and seamlessly integrate with the proposed pipeline.

The second part of this gap covers transformation into suitable formats (e.g., tensors), as well as normalisation and other standard techniques in ML. Each model typically follows a unique 'recipe' for these transformations, which is applied to data at both the training and inference steps. Generating ZKPs for these transformations will sometimes be quite involved: for example, images in training data that have attestations of provenance using C2PA may be in compressed formats (e.g. JPEG, PNG) that need to be decompressed before being used in a training algorithm. Filling this gap could simply involve extending ZKPoTs and ZKPoIs to ingest raw data, internally transforming and proving correctness of transformation as part of the larger proofs.

Xing et al. [61] examine this gap from the perspective of multiple independent parties involved in different tasks across the AI pipeline, where it is crucial for these parties to provide proofs of correctness to each other without disclosing additional information. In contrast, our proposed pipeline aims to offer verifiable guarantees even when all processes are conducted by a single actor, enabling auditors or users to independently verify these claims. Xing et al. divide data preprocessing into two sub-tasks: data collection and data refinement. Data collection involves selecting and acquiring data from original data owners based on specific criteria, which often requires querying databases. In the data refinement step, various preprocessing algorithms are applied enhance the features of the data, aiming to improve data quality. Xing et al. conclude that combining these dataset and preprocessing operations with ZKPs to validate correctness is rare. They find that the data collection process often aligns with the concept of Verifiable Databases (VDB), which, while not computationally intensive, face efficiency challenges due to high input-output complexity. Existing VDB solutions frequently use lightweight techniques such as digital signatures, polynomial commitments, or TEEs to improve efficiency.

Notably, Wang et al. [55] have designed the ezDPS scheme, which is the only work of which the authors are aware that proves the correctness of discrete wavelet transform (DWT) and principal component analysis (PCA) in ML scenarios. Elsewhere, Singh et al. [49] address decentralised AI pipelines, where different steps of the AI data processing and training pipeline are carried out by independent actors. The paper identifies the same lack of a unified AI pipeline for verifiability. The authors propose privacy-preserving ZKPs for basic operations such as aggregation, filtering, column sorting, and column inner-joining, but they do not address model-dependent transformations or proofs of training.

5.2 Gap between ZKPoT and ZKPoI

While a ZKPoT provides confidence that a model was honestly trained, it only confirms that there exists such a model. ZKPoTs are only useful to an end user receiving model outputs if they can be convinced they are being served the corresponding model and that the inference output was obtained from it. Similarly, while a ZKPoI provides a user with confidence that their input was honestly evaluated using some model, the proofs only confirm that there exists a model that produces a particular output when evaluated on the user's input. Additionally, ZKPoIs often take the arithmetised ML model as input, not the model itself. The user must therefore trust that the arithmetic circuit representation used for the ZKPoI is an 'honest' representation of the model. Linking between ZKPoI and ZKPoT closes this gap in verifiability and gives a user confidence that a service provider is serving the claimed model.

However, closing this gap is not necessarily straightforward. ZKPoI and ZKPoT schemes often use different, sometimes bespoke proof systems that are not directly compatible, e.g. quantisation is done differently. This is not an insurmountable challenge, but it has not been a research goal in most ZKPs for AI so far. The ZKAUDIT solution by Waiwitlikhit et al. [54] does indeed provide ZKPoTs and ZKPoIs that can be linked directly, showing that it is feasible. Clearly it is also possible to use a black-box approach to connect the proofs, where committed model weights are proven to correspond to model weights used in the ZKPoI through additional ZKPs, but it is not clear that this would be efficient.

6 Conclusions

In this paper, we proposed a framework for achieving end-to-end verifiability in AI pipelines, leveraging cryptographic tools such as digital signatures, cryptographic commitments, and Zero-Knowledge Proofs (ZKPs) to ensure strong guarantees of model correctness. Our focus was on verifiability of the whole end-to-end AI pipeline, including data extraction and refinement, model training, inference, and unlearning.

In §4 we mapped existing solutions and techniques onto the proposed framework, highlighting their potential contributions to enabling compliance with incoming AI regulation and for AI model deployment scenarios where a high level of assurance of the end-to-end AI pipeline is needed. In §5 we identified gaps in the AI pipeline that hinder full end-to-end verifiability, and highlighted research opportunities for bridging these gaps.

Our analysis has led us to following conclusions:

- The need for AI verifiability is a driver for developing ZKP standards. Research on ZKPs for AI is reaching a level where it can begin to be used in real-world scenarios to demonstrate compliance. However, there are currently no standard approaches for implementing them. Standards are needed both to enable interoperability of implementations of low-primitives, and also for conveying the cryptographic artefacts using common data formats.
- Regulators may need to play a critical role in enhancing trust in AI systems by verifying ZKPs. Since ZKPs do not necessarily hide algorithms such as the training process, regulators may need to verify proprietary model proofs on behalf of users to allow service providers to keep their model architectures private while giving strong cryptographic security guarantees. This approach addresses concerns regarding transparency in AI, particularly in light of the EU AI Act's requirements for auditability.
- There are a number of gaps in the verifiability of the AI pipeline. These gaps are possible focus areas for future research on full end-to-end verifiability.

Through the proposed framework, we hope to motivate the creation of robust, privacy-preserving AI systems that provide verifiable assurance across the entire AI pipeline, ultimately paving the way for more responsible and trustworthy AI deployment.

References

[1] 2000. Personal Information Protection and Electronic Documents Act (PIPEDA). Government of Canada, Justice Laws Website. https://laws-lois.justice.gc.ca/eng/acts/P-8.6/ Accessed: 2024-09-18.

[2] 2016. Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the Protection of Natural Persons with Regard to the Processing of Personal Data and on the Free Movement of Such Data (General Data Protection Regulation - GDPR). Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2016/679/oj Accessed: 2024-09-18.

[3] 2018. California Consumer Privacy Act of 2018 (CCPA). State of California, Department of Justice. https://oag.ca.gov/privacy/ccpa Accessed: 2024-09-18.

[4] 2024. *Coalition for Content Provenance and Authenticity (C2PA) Technical Specification v2.0.* Technical Report. Joint Development Foundation, Washington, DC, USA. https://c2pa.org/specifications/specifications/2.0/specs/C2PA_Specification.html Accessed: 2024-09-12.

[5] Kasra Abbaszadeh, Christodoulos Pappas, Jonathan Katz, and Dimitrios Papadopoulos. 2024. Zero-Knowledge Proofs of Training for Deep Neural Networks. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security* (Salt Lake City, UT, USA) *(CCS '24)*. Association for Computing Machinery, New York, NY, USA, 4316–4330. https://doi.org/10.1145/3658644.3670316

[6] Associated Press. 2024. Telling the Story - Standards. https://www.ap.org/about/news-values-and-principles/telling-the-story/. Accessed: 2024-09-13.

[7] Kar Balan, Andrew Gilbert, Alexander Black, Simon Jenni, Andy Parsons, and John Collomosse. 2023. DECORAIT - DECentralized Opt-in/out Registry for AI Training. In *Proceedings of the 20th ACM SIGGRAPH European Conference on Visual Media Production* (London, United Kingdom) *(CVMP '23)*. Association for Computing Machinery, New York, NY, USA, Article 4, 10 pages. https://doi.org/10.1145/3626495.3626506

[8] Kar Gabriel Balan, Andrew Gilbert, and John Collomosse. 2024. PDFed: Privacy-Preserving and Decentralized Asynchronous Federated Learning for Diffusion Models. In *Proceedings of 21st ACM SIGGRAPH Conference on Visual Media Production* (London, United Kingdom) *(CVMP '24)*. Association for Computing Machinery, New York, NY, USA, Article 8, 9 pages. https://doi.org/10.1145/3697294.3697306

[9] Eli Ben-Sasson, Iddo Bentov, Yinon Horesh, and Michael Riabzev. 2018. Scalable, transparent, and post-quantum secure computational integrity. *IACR Cryptol. ePrint Arch.* (2018), 46. http://eprint.iacr.org/2018/046

[10] Nir Bitansky, Alessandro Chiesa, Yuval Ishai, Omer Paneth, and Rafail Ostrovsky. 2013. Succinct non-interactive arguments via linear interactive proofs. In *Theory of Cryptography: 10th Theory of Cryptography Conference, TCC 2013, Tokyo, Japan, March 3-6, 2013. Proceedings*. Springer, 315–333.

[11] Camera Bits. 2024. Camera Bits Introduces Its Solution for Protecting Provenance of C2PA-Signed Photos in Effort to Help Combat Fake Imagery. https://home.camerabits.com/2024/05/06/press-release-camera-bits-introduces-its-solution-for-protecting-provenance-of-c2pa-signed-photos-in-effort-to-help-combat-fake-imagery/. Accessed: 2024-09-12.

[12] Jonathan Bootle, Alessandro Chiesa, Yuncong Hu, and Michele Orrù. 2022. Gemini: Elastic SNARKs for diverse environments. In *EUROCRYPT 2022, Part II (Lecture Notes in Computer Science (LNCS), Vol. 13276)*, Orr Dunkelman and Stefan Dziembowski (Eds.). Springer, Heidelberg, 427–457.

[13] Nicholas Carlini, Florian Tramèr, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Úlfar Erlingsson, Alina Oprea, and Colin Raffel. 2021. Extracting Training Data from Large Language Models. In *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, 2633–2650. https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting

[14] Bing-Jyue Chen, Suppakit Waiwitlikhit, Ion Stoica, and Daniel Kang. 2024. ZKML: An Optimizing System for ML Inference in Zero-Knowledge Proofs. In *Proceedings of the Nineteenth European Conference on Computer Systems* (Athens, Greece) *(EuroSys '24)*. Association for Computing Machinery, New York, NY, USA, 560–574. https://doi.org/10.1145/3627703.3650088

[15] John Collomosse and Andy Parsons. 2024. To Authenticity, and Beyond! Building Safe and Fair Generative AI Upon the Three Pillars of Provenance. *IEEE Computer Graphics and Applications* 44, 03 (may 2024), 82–90. https://doi.org/10.1109/MCG.2024.3380168

[16] Trisha Datta, Binyi Chen, and Dan Boneh. 2024. VerITAS: Verifying Image Transformations at Scale. Cryptology ePrint Archive, Paper 2024/1066. https://eprint.iacr.org/2024/1066

[17] Boxiang Dong, Bo Zhang, and Hui (Wendy) Wang. 2021. VeriDL: Integrity Verification of Outsourced Deep Learning Services. In *Machine Learning and Knowledge Discovery in Databases. Research Track: European Conference, ECML PKDD 2021, Bilbao, Spain, September 13–17, 2021, Proceedings, Part II* (Bilbao, Spain). Springer-Verlag, Berlin, Heidelberg, 583–598. https://doi.org/10.1007/978-3-030-86520-7_36

[18] Thorsten Eisenhofer, Doreen Riepel, Varun Chandrasekaran, Esha Ghosh, Olga Ohrimenko, and Nicolas Papernot. 2025. Verifiable and Provably Secure Machine Unlearning. In *Proceedings of the 3rd IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*. https://arxiv.org/abs/2210.09126 To appear.

[19] Jens Ernstberger, Stefanos Chaliasos, Liyi Zhou, Philipp Jovanovic, and Arthur Gervais. 2024. Do You Need a Zero Knowledge Proof? *Cryptology ePrint Archive* (2024).

[20] European Data Protection Board. 2024. Opinion 28/2024 on Certain Data Protection Aspects Related to the Processing of Personal Data in the Context of AI Models. https://www.edpb.europa.eu/system/files/2024-12/edpb_opinion_202428_ai-models_en.pdf Adopted on 17 December 2024.

[21] Olive Franzese, Ali Shahin Shamsabadi, and Hamed Haddadi. 2024. OATH: Efficient and Flexible Zero-Knowledge Proofs of End-to-End ML Fairness. arXiv:2410.02777 [cs.CY] https://arxiv.org/abs/2410.02777

[22] Ji Gao, Sanjam Garg, Mohammad Mahmoody, and Prashant Nalini Vasudevan. 2022. Deletion Inference, Reconstruction, and Compliance in Machine (Un)Learning. *Proceedings on Privacy Enhancing Technologies* 2022, 3 (2022), 415–436. https://doi.org/10.56553/popets-2022-0079

[23] Sanjam Garg, Aarushi Goel, Somesh Jha, Saeed Mahloujifar, Mohammad Mahmoody, Guru-Vamsi Policharla, and Mingyuan Wang. 2023. Experimenting with Zero-Knowledge Proofs of Training. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security* (Copenhagen, Denmark) *(CCS '23)*. Association for Computing Machinery, New York, NY, USA, 1880–1894. https://doi.org/10.1145/3576915.3623202

[24] Shafi Goldwasser, Huijia Lin, and Aviad Rubinstein. 2011. Delegation of computation without rejection problem from designated verifier CS-proofs. *Cryptology ePrint Archive* (2011).

[25] Feng Hao. 2017. Schnorr Non-interactive Zero-Knowledge Proof. RFC 8235. https://doi.org/10.17487/RFC8235

[26] Chenyu Huang, Jianzong Wang, Huangxun Chen, Shijing Si, Zhangcheng Huang, and Jing Xiao. 2022. zkMLaaS: a Verifiable Scheme for Machine Learning as a Service. In *GLOBECOM 2022 - 2022 IEEE Global Communications Conference*. 5475–5480. https://doi.org/10.1109/GLOBECOM48099.2022.10000784

[27] IBM. 2024. What is fine-tuning? https://www.ibm.com/topics/fine-tuning

[28] Daniel Kang, Kobi Gurkan, and Anna Rose. 2023. Fighting AI-Generated Audio with Attested Microphones and ZK-SNARKs: The Attested Audio Experiment. https://medium.com/@danieldkang/fighting-ai-generated-audio-with-attested-microphones-and-zk-snarks-the-attested-audio-experiment-d6ea0fc296ac Accessed: 2024-09-16.

[29] Daniel Kang, Tatsunori Hashimoto, Ion Stoica, and Yi Sun. 2022. ZK-IMG: Attested Images via Zero-Knowledge Proofs to Fight Disinformation. arXiv:2211.04775 [cs.CR] https://arxiv.org/abs/2211.04775

[30] Daniel Kang, Tatsunori Hashimoto, Ion Stoica, and Yi Sun. 2023. Scaling up Trustless DNN Inference with Zero-Knowledge Proofs. In *NeurIPS RegulateML Workshop*. arXiv:2210.08674 [cs.CR] https://arxiv.org/abs/2210.08674

[31] Hankyung Ko, Ingeun Lee, Seunghwa Lee, Jihye Kim, and Hyunok Oh. 2021. Efficient Verifiable Image Redacting based on zk-SNARKs. In *Proceedings of the 2021 ACM Asia Conference on Computer and Communications Security* (Virtual Event, Hong Kong) *(ASIA CCS '21)*. Association for Computing Machinery, New York, NY, USA, 213–226. https://doi.org/10.1145/3433210.3453110

[32] Qi Li, Zhuotao Liu, Qi Li, and Ke Xu. 2023. martFL: Enabling Utility-Driven Data Marketplace with a Robust and Verifiable Federated Learning Architecture. In *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security* (Copenhagen, Denmark) *(CCS '23)*. Association for Computing Machinery, New York, NY, USA, 1496–1510. https://doi.org/10.1145/3576915.3623134

[33] Tianyi Liu, Xiang Xie, and Yupeng Zhang. 2021. zkCNN: Zero knowledge proofs for convolutional neural network predictions and accuracy. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS 2021)*, Giovanni Vigna and Elaine Shi (Eds.). ACM Press, 2968–2985.

[34] Shayne Longpre, Robert Mahari, Naana Obeng-Marnu, William Brannon, Tobin South, Jad Kabbara, and Sandy Pentland. 2024. Data Authenticity, Consent, and Provenance for AI Are All Broken: What Will It Take to Fix Them? *An MIT Exploration of Generative AI* (March 2024). Available at: https://mit-genai.pubpub.org/pub/uk7op8zs.

[35] Philipp Lüthi, Thibault Gagnaux, and Marcel Gygli. 2020. Distributed Ledger for Provenance Tracking of Artificial Intelligence Assets. In *Privacy and Identity Management. Data for Better Living: AI and Privacy*. IFIP Advances in Information and Communication Technology, Vol. 576. Springer International Publishing, 411–426. https://doi.org/10.1007/978-3-030-42504-3_26

[36] Juan Ma, Hao Liu, Mingyue Zhang, and Zhiming Liu. 2024. VPFL: Enabling verifiability and privacy in federated learning with zero-knowledge proofs. *Know.-Based Syst.* 299, C (Sept. 2024), 15 pages. https://doi.org/10.1016/j.knosys.2024.112115

[37] Brendan McMahan and Daniel Ramage. 2017. Federated learning: Collaborative machine learning without centralized training data. *Google Research Blog* 3 (2017). https://research.google/blog/federated-learning-collaborative-machine-learning-without-centralized-training-data

[38] Silvio Micali. 2000. Computationally Sound Proofs. *SIAM J. Comput.* 30, 4 (Oct. 2000), 1253–1298. https://doi.org/10.1137/S0097539795284959

[39] Modulus Labs. 2023. The cost of intelligence: Proving machine learning inference with zero-knowledge. Available at: https://drive.google.com/file/d/1tylpowpaqcOhKQtYolPlqvx6R2Gv4IzE/view.

[40] Antonio Muñoz, Ruben Ríos, Rodrigo Román, and Javier López. 2023. A survey on the (in)security of trusted execution environments. *Comput. Secur.* 129, C (June 2023), 26 pages. https://doi.org/10.1016/j.cose.2023.103180

[41] Assa Naveh and Eran Tromer. 2016. PhotoProof: Cryptographic Image Authentication for Any Set of Permissible Transformations. In *2016 IEEE Symposium on Security and Privacy (SP)*. 255–271. https://doi.org/10.1109/SP.2016.23

[42] Thanh Tam Nguyen, Thanh Trung Huynh, Zhao Ren, Phi Le Nguyen, Alan Wee-Chung Liew, Hongzhi Yin, and Quoc Viet Hung Nguyen. 2024. A Survey of Machine Unlearning. arXiv:2209.02299 [cs.LG] https://arxiv.org/abs/2209.02299

[43] OpenAI. 2024. C2PA in DALL·E 3. https://help.openai.com/en/articles/8912793-c2pa-in-dall-e-3. Accessed: 2024-09-12.

[44] European Parliament and Council. 2024. Regulation (EU) 2024/1689 on laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) and amending certain Union legislative acts. https://eur-lex.europa.eu/eli/reg/2024/1689/oj. Official Journal of the European Union, L 257, 25.07.2024, p. 1–103.

[45] M. O. Rabin. 1979. *DIGITALIZED SIGNATURES AND PUBLIC-KEY FUNCTIONS AS INTRACTABLE AS FACTORIZATION.* Technical Report. USA.

[46] Timon Rückel, Johannes Sedlmeir, and Peter Hofmann. 2022. Fairness, integrity, and privacy in a scalable blockchain-based federated learning system. *Computer Networks* 202 (2022), 108621. https://doi.org/10.1016/j.comnet.2021.108621

[47] Srinath Setty. 2020. Spartan: Efficient and General-Purpose zkSNARKs Without Trusted Setup. In *Advances in Cryptology – CRYPTO 2020*, Daniele Micciancio and Thomas Ristenpart (Eds.). Springer International Publishing, Cham, 704–737.

[48] Ali Shahin Shamsabadi, Sierra Calanda Wyllie, Nicholas Franzese, Natalie Dullerud, Sébastien Gambs, Nicolas Papernot, Xiao Wang, and Adrian Weller. 2023. Confidential-PROFITT: Confidential PROof of FaIr Training of Trees. In *The Eleventh International Conference on Learning Representations*. https://openreview.net/forum?id=iIfDQVyuFD

[49] Nitin Singh, Pankaj Dayama, and Vinayaka Pandit. 2022. Zero Knowledge Proofs Towards Verifiable Decentralized AI Pipelines. In *Financial Cryptography and Data Security: 26th International Conference, FC 2022, Grenada, May 2–6, 2022, Revised Selected Papers* (Grenada, Grenada). Springer-Verlag, Berlin, Heidelberg, 248–275. https://doi.org/10.1007/978-3-031-18283-9_12

[50] Sony. 2024. Sony Delivers Highly Anticipated Firmware Updates Including C2PA Compliancy and Ensuring Authenticity of Images. https://www.sony.co.uk/presscentre/sony-delivers-highly-anticipated-firmware-updates-including-c2pa-compliancy-and-ensuring-authenticity-of-images. Accessed: 2024-09-12.

[51] Tobin South, Alexander Camuto, Shrey Jain, Shayla Nguyen, Robert Mahari, Christian Paquin, Jason Morton, and Alex 'Sandy' Pentland. 2024. Verifiable evaluations of machine learning models using zkSNARKs. arXiv:2402.02675 [cs.LG] https://arxiv.org/abs/2402.02675

[52] Haochen Sun, Tonghe Bai, Jason Li, and Hongyang Zhang. 2025. zkDL: Efficient Zero-Knowledge Proofs of Deep Learning Training. *Trans. Info. For. Sec.* 20 (Jan. 2025), 914–927. https://doi.org/10.1109/TIFS.2024.3520863

[53] Paul Valiant. 2008. Incrementally Verifiable Computation or Proofs of Knowledge Imply Time/Space Efficiency. In *Theory of Cryptography*, Ran Canetti (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 1–18.

[54] Suppakit Waiwitlikhit, Ion Stoica, Yi Sun, Tatsunori Hashimoto, and Daniel Kang. 2024. Trustless audits without revealing data or models. In *Proceedings of the 41st International Conference on Machine Learning* (Vienna, Austria) *(ICML'24)*. JMLR.org, Article 2036, 14 pages.

[55] Haodi Wang and Thang Hoang. 2023. ezDPS: An Efficient and Zero-Knowledge Machine Learning Inference Pipeline. *Proceedings on Privacy Enhancing Technologies* 2023, 2 (2023). https://doi.org/10.56553/popets-2023-0061

[56] Lingling Wang, Xueqin Zhao, Zhongkai Lu, Lin Wang, and Shouxun Zhang. 2023. Enhancing privacy preservation and trustworthiness for decentralized federated learning. *Information Sciences* 628 (2023), 449–468. https://doi.org/10.1016/j.ins.2023.01.130

[57] Zhipeng Wang, Nanqing Dong, Jiahao Sun, William Knottenbelt, and Yike Guo. 2025. zkFLzkFL: Zero-Knowledge Proof-Based Gradient Aggregation for Federated Learning. *IEEE Transactions on Big Data* 11, 2 (2025), 447–460. https://doi.org/10.1109/TBDATA.2024.3403370

[58] Chenkai Weng, Kang Yang, Xiang Xie, Jonathan Katz, and Xiao Wang. 2021. Mystique: Efficient Conversions for Zero-Knowledge Proofs with Applications to Machine Learning. In *30th USENIX Security Symposium (USENIX Security 21)*. USENIX Association, 501–518. https://www.usenix.org/conference/usenixsecurity21/presentation/weng

[59] Jiasi Weng, Shenglong Yao, Yuefeng Du, Junjie Huang, Jian Weng, and Cong Wang. 2024. Proof of Unlearning: Definitions and Instantiation. *IEEE Transactions on Information Forensics and Security* 19 (2024), 3309–3323. https://doi.org/10.1109/TIFS.2024.3358993

[60] Herbert Woisetschläger, Simon Mertel, Christoph Krönke, Ruben Mayer, and Hans-Arno Jacobsen. 2024. Federated Learning and AI Regulation in the European Union: Who is Liable? – An Interdisciplinary Analysis. Presented at Generative AI + Law (GenLaw) '24, ICML 2024, available at https://blog.genlaw.org/pdfs/genlaw_icml2024/16.pdf.

[61] Zhibo Xing, Zijian Zhang, Jiamou Liu, Ziang Zhang, Meng Li, Liehuang Zhu, and Giovanni Russello. 2023. Zero-knowledge Proof Meets Machine Learning in Verifiability: A Survey. arXiv:2310.14848 [cs.LG] https://arxiv.org/abs/2310.14848

[62] Jie Xu, Zihan Wu, Cong Wang, and Xiaohua Jia. 2024. Machine Unlearning: Solutions and Challenges. *IEEE Transactions on Emerging Topics in Computational Intelligence* 8, 3 (June 2024), 2150–2168. https://doi.org/10.1109/tetci.2024.3379240

[63] Chhavi Yadav, Amrita Roy Chowdhury, Dan Boneh, and Kamalika Chaudhuri. 2024. FairProof : Confidential and Certifiable Fairness for Neural Networks. In *Proceedings of the 41st International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 235)*, Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (Eds.). PMLR, 55682–55705. https://proceedings.mlr.press/v235/yadav24a.html

[64] Eva Zhang, Akilesh Potti, and Micah Goldblum. 2024. vTune: Verifiable Fine-Tuning Through Backdooring. In *NeurIPS Workshop on Statistical Frontiers in LLMs and Foundation Models*. arXiv:2411.06611 [cs.LG] https://arxiv.org/abs/2411.06611

[65] Justin Zhao, Timothy Wang, Wael Abid, Geoffrey Angus, Arnav Garg, Jeffery Kinnison, Alex Sherstinsky, Piero Molino, Travis Addair, and Devvret Rishi. 2024. LoRA Land: 310 Fine-tuned LLMs that Rival GPT-4, A Technical Report. arXiv:2405.00732 [cs.CL] https://arxiv.org/abs/2405.00732

[66] Lingchen Zhao, Qian Wang, Cong Wang, Qi Li, Chao Shen, and Bo Feng. 2021. VeriML: Enabling Integrity Assurances and Fair Payments for Machine Learning as a Service. *IEEE Transactions on Parallel and Distributed Systems* 32, 10 (2021), 2524–2540. https://doi.org/10.1109/TPDS.2021.3068195